

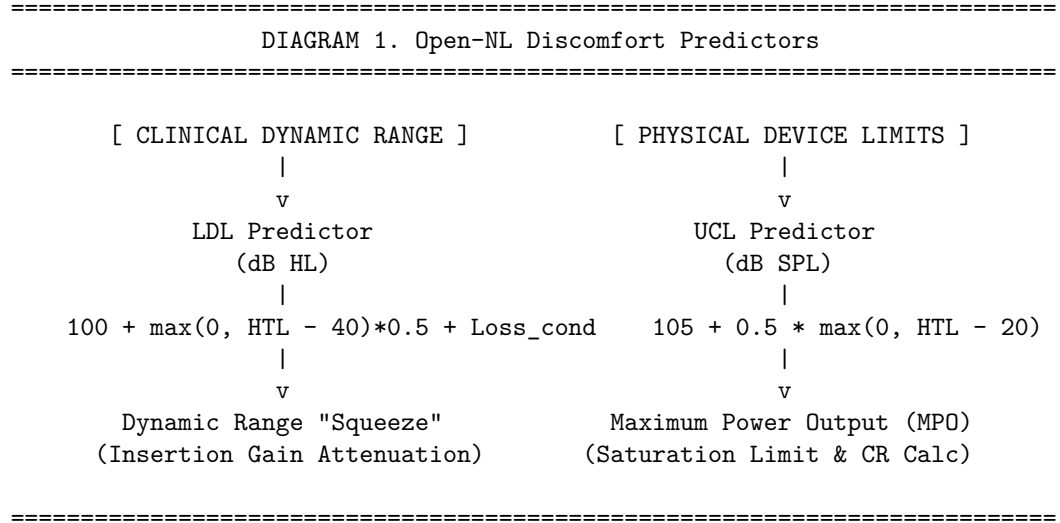
## Supplementary Material: Open-NL Algorithmic Pipeline and Complete Parameter Specification

This supplementary document provides the exact mathematical formulations, closed-form piecewise functions, and complete numerical parameter specifications governing the twelve internal processing stages of the Open-NL algorithmic cascade.

### C. The Twelve-Stage Execution Cascade

Open-NL operates as a multi-stage parameterized shape generator. Rather than relying on static compiled lookup tables, Open-NL calculates target insertion gains dynamically through a series of twelve explicitly defined, cascaded mathematical modules. Each step in the gain derivation process is exposed natively in R, available for researchers to inspect, modify, and tune.

*Terminology Note:* Throughout this framework, the algorithm utilizes two distinct discomfort predictors for different theoretical purposes: an HL-domain “LDL” (Loudness Discomfort Level) predictor used for estimating clinical audiometric dynamic range (Stage 8), and an SPL-domain “UCL” (Uncomfortable Loudness Level) predictor for establishing physical device saturation limits (Stage 11). A visual flowchart mapping each predictor to its downstream algorithm function is provided in Diagram 1.



The twelve modules operate in a strictly defined cascaded execution order to prevent unintended interactions between additive boosters and soft limiters:

1. **Stage 1: Conductive Component Separation & Dynamic Range Baseline** (Section II.D)

2. **Stage 2: Decoupled Half-Gain Base Anchor Calculation** (Section II.E)
  3. **Stage 3: Experience-Level Shaping & Log-Frequency  $C_{vals}$  Interpolation** (Section II.F)
  4. **Stage 4: Reverse-Slope Low-Frequency Attenuation Floor** (Section II.G)
  5. **Stage 5: Slope-Dependent Low-Frequency Penalty (SD-LFP) with Profound HF Bypass** (Section II.H)
  6. **Stage 6: Severe-Loss Audibility Booster** (Section II.I)
  7. **Stage 7: Soft-Compression High-Frequency Desensitization** (Section II.J)
  8. **Stage 8: Dynamic Range Mapping (LDL Squeeze)** (Section II.K)
  9. **Stage 9: Transducer Bandwidth Roll-off** (Section II.L)
  10. **Stage 10: Multi-Channel WDRC Mapping, Input/Output Pivot, and Demographic Adjustments** (Section II.M)
  11. **Stage 11: Acoustic Venting, Coupling Loss, and Receiver Saturation Limits** (Section II.N)
  12. **Stage 12: Embedded Nelder-Mead Simplex Optimization & Physiological Loudness Ceilings** (Section II.O)
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## D. Stage 1: Conductive Component Separation & Dynamic Range Baseline

The algorithm first decomposes total hearing threshold levels (HTL) into their physiological constituents: the sensorineural threshold ( $HTL_{sn}$ ) and the conductive component ( $Loss_{cond}$ , corresponding to the air-bone gap, ABG):

$$Loss_{cond}(f) = ABG(f) \quad (1)$$

$$HTL_{sn}(f) = \max(0, HTL(f) - Loss_{cond}(f)) \quad (2)$$

To ensure that equal-loudness reshaping penalties apply only to sensorineural pathology, the algorithm computes a frequency-specific sensorineural ratio  $R_{sn}(f)$ :

$$R_{sn}(f) = \begin{cases} \frac{HTL_{sn}(f)}{HTL(f)}, & \text{if } HTL(f) > 0 \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Pure conductive losses ( $Loss_{cond} = HTL$ ) have normal outer and inner hair cell functioning and intact basilar membrane mechanics; thus,  $R_{sn} = 0$ , bypassing cochlear recruitment and equal-loudness correction arrays.

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## E. Stage 2: Decoupled Half-Gain Base Anchor Calculation

The foundational WDRC anchor for average conversational speech (65 dB SPL) is derived using a frequency-specific adaptation of the half-gain rule (Lybarger, 1944). Unlike linear prescriptions that anchor to a broadband Pure Tone Average (PTA), this anchor is decoupled frequency-by-frequency:

$$G_{base}(f) = \alpha \cdot \text{HTL}_{sn}(f) + C_{interp}(f) \quad (4)$$

where  $\alpha = 0.46$  is the nominal half-gain multiplier (loosely derived from Byrne & Dillon, 1986), and  $C_{interp}(f)$  is the frequency-shaping correction array derived in Stage 3.

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## F. Stage 3: Experience-Level Shaping & Log-Frequency $C_{vals}$ Interpolation

To account for listener acclimatization and preferred listening levels (Keidser et al., 2012), Open-NL modulates the frequency-shaping array  $C_{interp}$  across eight discrete anchor frequencies:

$$\mathbf{f}_c = [250, 500, 1000, 2000, 3000, 4000, 6000, 8000] \text{ Hz} \quad (5)$$

The discrete shaping vectors  $\mathbf{C}$  are defined as: - **Experienced Users** (standard baseline):

$$\mathbf{C}_{exp} = [-8, -1, 3, 1, 0, 0, 0, 0] \text{ dB} \quad (6)$$

- **New Users** (nominal  $-3$  dB reduction to combat occlusion and sharpness):

$$\mathbf{C}_{new} = \mathbf{C}_{exp} - 3 = [-11, -4, 0, -2, -3, -3, -3, -3] \text{ dB} \quad (7)$$

- **Power Users** (prioritizes raw audibility over acoustic comfort):

$$\mathbf{C}_{power} = \mathbf{C}_{exp} + 3 = [-5, +2, 6, 4, 3, 3, 3, 3] \text{ dB} \quad (8)$$

For any calculation frequency  $f$ ,  $C_{interp}(f)$  is evaluated via piecewise log-linear interpolation across  $\mathbf{f}_c$  and scaled by the sensorineural proportion  $R_{sn}(f)$ :

$$C_{interp}(f) = \text{interp}_{\log_{10}}(f, \mathbf{f}_c, \mathbf{C}) \cdot R_{sn}(f) \quad (9)$$


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## G. Stage 4: Reverse-Slope Low-Frequency Attenuation Floor

For reverse-slope configurations (where low frequencies are significantly worse than high frequencies), attempting to fully restore low-frequency audibility risks upward spread of masking, where high-energy low-frequency vowels mask low-energy high-frequency consonants. The algorithm calculates the low-to-high sensorineural threshold difference:

$$\text{Diff}_{rs} = \max(0, \overline{\text{HTL}}_{sn, \leq 500} - \overline{\text{HTL}}_{sn, \geq 2000}) \quad (10)$$

where  $\overline{\text{HTL}}_{sn, \leq 500}$  is the mean threshold across  $f \leq 500$  Hz, and  $\overline{\text{HTL}}_{sn, \geq 2000}$  is the mean threshold across  $f \geq 2000$  Hz. If  $\text{Diff}_{rs} > 15$  dB, a transition factor  $RS_{factor} \in [0, 1]$  is computed:

$$RS_{factor} = \max\left(0, \min\left(1, \frac{\text{Diff}_{rs} - 15}{20}\right)\right) \quad (11)$$

A log-linear low-frequency taper  $W_{LF}(f)$  is applied below 1000 Hz:

$$W_{LF}(f) = \max\left(0, \min\left(1, 1 - \frac{\log_{10}(f) - \log_{10}(250)}{\log_{10}(1000/250)}\right)\right) \quad (12)$$

Across octave bands,  $W_{LF}(f)$  evaluates to 1.0 at 250 Hz, 0.5 at 500 Hz, and 0.0 at  $f \geq 1000$  Hz. The frequency-shaping array is then dynamically flattened toward a  $-10$  dB floor:

$$C'_{interp}(f) = C_{interp}(f) \cdot (1 - RS_{factor} \cdot W_{LF}(f)) + (-10 \cdot RS_{factor} \cdot W_{LF}(f)) \quad (13)$$

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## H. Stage 5: Slope-Dependent Low-Frequency Penalty (SD-LFP) with Profound HF Bypass

For sloping high-frequency losses, applying excessive low-frequency gain causes normal low frequencies to dominate broadband loudness. Open-NL computes the high-frequency slope:

$$\text{Slope} = \max(0, \overline{\text{HTL}}_{sn, \geq 2000} - \overline{\text{HTL}}_{sn, \leq 500}) \quad (14)$$

To operationalize the empirical findings of Byrne, Parkinson, and Newall (1990)—who showed that listeners with high-frequency losses exceeding 70–95 dB HL

require low-frequency speech cues—Open-NL incorporates a **Profound High-Frequency Bypass**:

$$PF_{bypass} = \max \left( 0, \min \left( 1, \frac{95 - \overline{\text{HTL}}_{sn, \geq 2000}}{25} \right) \right) \quad (15)$$

The low-frequency penalty is scaled over a 20 dB slope window, capped at 15 dB:

$$\text{LF}_{penalty} = \max \left( 0, \min \left( 1, \frac{\text{Slope} - 15}{20} \right) \right) \cdot 15 \cdot PF_{bypass} \quad (16)$$

The penalty is then tapered below 1000 Hz using  $W_{LF}(f)$ :

$$C''_{interp}(f) = C'_{interp}(f) - (\text{LF}_{penalty} \cdot W_{LF}(f)) \quad (17)$$

The resulting baseline target at 65 dB SPL is:

$$G_{65}^{(0)}(f) = 0.46 \cdot \text{HTL}_{sn}(f) + C''_{interp}(f) \quad (18)$$

#### Ablation of SD-LFP Constraints

Table I provides the reference audiometric profiles (A1–A7, Johnson & Dillon, 2011). Table II demonstrates the mechanical impact of the SD-LFP constraint on the initial heuristic seeds.

**TABLE I. Reference Audiometric Profiles (Adapted from Johnson & Dillon, 2011).** *Thresholds are in dB HL. Profiles A1–A5 are purely sensorineural. A6 is mixed (30 dB ABG). A7 is conductive (50 dB ABG).*

| Profile   | Type                       | 250<br>Hz | 500<br>Hz | 1000<br>Hz | 2000<br>Hz | 4000<br>Hz | 8000<br>Hz | ABG  |
|-----------|----------------------------|-----------|-----------|------------|------------|------------|------------|------|
| <b>A1</b> | Flat<br>mod-<br>erate      | 15        | 20        | 30         | 40         | 50         | 60         | 0 dB |
| <b>A2</b> | Reverse<br>slope           | 60        | 50        | 40         | 30         | 20         | 15         | 0 dB |
| <b>A3</b> | Moderately<br>slop-<br>ing | 40        | 20        | 40         | 50         | 55         | 60         | 0 dB |
| <b>A4</b> | Steeply<br>slop-<br>ing    | 0         | 0         | 10         | 40         | 70         | 80         | 0 dB |

| Profile   | Type                    | 250<br>Hz | 500<br>Hz | 1000<br>Hz | 2000<br>Hz | 4000<br>Hz | 8000<br>Hz | ABG   |
|-----------|-------------------------|-----------|-----------|------------|------------|------------|------------|-------|
| <b>A5</b> | Steeply<br>slop-<br>ing | 10        | 10        | 20         | 60         | 80         | 100        | 0 dB  |
| <b>A6</b> | Mixed                   | 50        | 55        | 60         | 65         | 75         | 80         | 30 dB |
| <b>A7</b> | Conductive              | 50        | 50        | 50         | 50         | 50         | 50         | 50 dB |

**TABLE II. Ablation of SD-LFP Constraints on Optimizer Seed (65 dB SPL Input).**

| Profile             | Unconstrained<br>Seed SII | Unconstrained<br>Seed Sones | SD-LFP Seed<br>SII | SD-LFP Seed<br>Sones |
|---------------------|---------------------------|-----------------------------|--------------------|----------------------|
| A1 (Flat<br>mod)    | 0.86                      | 7.2                         | 0.86               | 7.2                  |
| A2 (Reverse)        | 0.88                      | 6.3                         | 0.88               | 6.0                  |
| A3 (Mod<br>sloping) | 0.79                      | 6.8                         | 0.79               | 6.8                  |
| A4 (Steeply)        | 0.81                      | 6.9                         | 0.81               | 6.9                  |
| A5 (Steeply)        | 0.68                      | 7.1                         | 0.68               | 6.0                  |
| A6 (Mixed)          | 0.82                      | 3.0                         | 0.82               | 3.0                  |
| A7<br>(Conductive)  | 0.97                      | 1.0                         | 0.97               | 1.0                  |

## I. Stage 6: Severe-Loss Audibility Booster

To overcome inner hair cell loss in severe impairments, an opt-in, bounded **Severe-Loss Booster** can be applied:

$$G_{slb}(f) = B_{en} \cdot 0.15 \cdot \max(0, \min(80, \text{HTL}_{sn}(f)) - T_{onset}) \quad (19)$$

where  $B_{en} \in \{0, 1\}$  (default: 0, disabled), and  $T_{onset} = 70$  dB HL (conservative default) or 60 dB HL (aggressive ablation mode). The intermediate target is:

$$G_{65}^{(1)}(f) = G_{65}^{(0)}(f) + G_{slb}(f) \quad (20)$$

## J. Stage 7: Soft-Compression High-Frequency Desensitization

To prevent unconstrained audibility maximization from prescribing intolerable high-frequency gain in steeply sloping losses, Open-NL applies a dynamic soft-compression envelope ( $L_{gain}$ ):

$$L_{gain}(f) = 30 + 0.4 \cdot \max(0, \text{HTL}_{sn}(f) - 60) \quad (21)$$

(Note: For patients in “Moderate” or “High” distortion categories,  $L_{gain}$  is reduced by 10 dB).

Excess gain above this dynamic limit is:

$$\text{Excess}(f) = \max\left(0, G_{65}^{(1)}(f) - L_{gain}(f)\right) \quad (22)$$

The sloping factor  $S_{factor}(f)$  relative to the best low-frequency threshold ( $\text{HTL}_{bestlow} = \min_{f' \leq 1000} \text{HTL}_{sn}(f')$ ) and the high-frequency fade-in weight  $W_{hf}(f)$  are:

$$S_{factor}(f) = \max\left(0, \min\left(1, \frac{\text{HTL}_{sn}(f) - \text{HTL}_{bestlow} - 25}{20}\right)\right) \quad (23)$$

$$W_{hf}(f) = \max\left(0, \min\left(1, \frac{f - 2000}{2000}\right)\right) \quad (24)$$

Applying 2:1 soft compression to the excess yields:

$$G_{65}^{(2)}(f) = G_{65}^{(1)}(f) - (W_{hf}(f) \cdot S_{factor}(f) \cdot 0.50 \cdot \text{Excess}(f)) \quad (25)$$

If explicit cochlear dead regions ( $f_{e\_hf}$ ,  $f_{e\_lf}$ ) or distortion categories (Margolis et al., 2025) are defined, steep parametric roll-offs are applied:

$$G_{65}^{(2)}(f) \leftarrow G_{65}^{(2)}(f) - \max\left(0, \log_2\left(\frac{f}{1.7f_{e\_hf}}\right)\right) \cdot 30 - \max\left(0, \log_2\left(\frac{0.57f_{e\_lf}}{f}\right)\right) \cdot 30 \quad (26)$$

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## K. Stage 8: Dynamic Range Mapping (LDL Squeeze)

To accommodate reduced clinical dynamic ranges (DSL v5.0 philosophy; Scollie et al., 2005), Open-NL predicts an HL-domain Loudness Discomfort Level:

$$\text{LDL}_{pred}(f) = 100 + 0.5 \cdot \max(0, \text{HTL}_{sn}(f) - 40) + \text{Loss}_{cond}(f) \quad (27)$$

When measured  $\text{LDL}_{\text{meas}}(f)$  is lower than predicted, the dynamic range discrepancy  $\Delta_{\text{LDL}}(f)$  is quantified:

$$\Delta_{\text{LDL}}(f) = \text{LDL}_{\text{meas}}(f) - \text{LDL}_{\text{pred}}(f) \quad (28)$$

Target gain is attenuated by 0.2 dB per dB of dynamic range squeeze:

$$G_{65}^{(3)}(f) = G_{65}^{(2)}(f) + (0.2 \cdot \Delta_{\text{LDL}}(f)) \quad (29)$$

Simultaneously, the baseline compression ratio increases by +0.02 per dB of squeeze (Stage 10).

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### L. Stage 9: Transducer Bandwidth Roll-off

Acoustic transducers physically struggle to reproduce frequencies at the extremes of the spectrum ( $\leq 250$  Hz and  $\geq 6000$  Hz), where massive gain leads to distortion and feedback. Open-NL applies a continuous fractional bandwidth roll-off multiplier  $M_{bw}(f)$  defined over seven anchor frequencies:

$$\mathbf{f}_{bw} = [250, 500, 1000, 2000, 4000, 6000, 8000] \text{ Hz} \quad (30)$$

$$\mathbf{w}_{bw} = [0.7, 1.0, 1.0, 1.0, 1.0, 0.8, 0.5] \quad (31)$$

The closed-form continuous multiplier is obtained via piecewise log-linear interpolation:

$$M_{bw}(f) = \text{interp}_{\log_{10}}(f, \mathbf{f}_{bw}, \mathbf{w}_{bw}) \quad (32)$$

$$G_{65}^{(4)}(f) = G_{65}^{(3)}(f) \cdot M_{bw}(f) \quad (33)$$


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### M. Stage 10: Multi-Channel WDRC Mapping, Input/Output Pivot, and Demographic Adjustments

Target gain at arbitrary overall input level  $L_{in}$  (e.g., 50, 65, 80 dB SPL) is computed using a multi-channel wide dynamic range compression architecture pivoted around the Long-Term Average Speech Spectrum (LTASS).

1. **Speech Spectrum Pivot:** Let  $P(f)$  be the critical-band normal speech level at 65 dB SPL overall. The band-specific input level is:

$$L_{band}(f) = P(f) + (L_{in} - 65) \quad (34)$$

## 2. Dynamic Compression Ratio Calculation:

$$\text{CR}_{base}(f) = 1 + \frac{\max(0, \text{HTL}_{sn}(f) - 20)}{40} - (0.02 \cdot \Delta_{LDL}(f)) \quad (35)$$

For severe loss ( $\text{HTL}_{sn} > 65$  dB HL), compression reduces back toward linear to preserve the temporal speech envelope (Keidser et al., 2007):

$$W_{freq}(f) = \max\left(0, \min\left(1, \frac{f - 500}{2500}\right)\right) \quad (36)$$

$$\text{CR}_{loud}(f) = \text{CR}_{base}(f) - \left(\frac{\max(0, \text{HTL}_{sn}(f) - 65)}{30} \cdot (1.5 - 0.5 \cdot W_{freq}(f))\right) \quad (37)$$

$\text{CR}_{loud}(f)$  is clamped between 1.0 and a maximum  $\text{CR}_{max}(f) = 1.5 + 0.9 \cdot W_{freq}(f)$  (further relaxed for age  $> 60$ ). In Comfort in Noise (CIN) mode,  $\text{CR}_{loud}$  is clamped to  $\leq 1.5$ .

*(Note on Evidentiary Asymmetry: The 1.5:1 CIN clamp is an asserted engineering heuristic inspired by NAL-NL3 to curb listening fatigue in noise, lacking direct empirical derivation. In contrast, the global upper ceiling on compression ratios ( $\leq 3.0 : 1$ ) is directly supported by extensive psychoacoustic literature [Souza, 2002; Souza et al., 2006], which demonstrates that compression ratios exceeding 3.0:1 cause severe temporal envelope flattening and acoustic contrast degradation).*

3. **Variable Compression Threshold (CT):** Overall CT ( $\text{CT}_{overall}$ ) scales from 30 to 45 dB SPL as a function of threshold. The band-level threshold is  $\text{CT}_{band}(f) = P(f) + (\text{CT}_{overall} - 65)$  (reduced by 10 dB in CIN mode).

4. **Piecewise Linear-Compressive Spline:** Gain at the compression threshold is:

$$G_{CT}(f) = \begin{cases} G_{65}^{(4)}(f), & \text{if } \text{CT}_{band}(f) > P(f) \\ G_{65}^{(4)}(f) + (P(f) - \text{CT}_{band}(f)) \left(1 - \frac{1}{\text{CR}_{loud}(f)}\right), & \text{if } \text{CT}_{band}(f) \leq P(f) \end{cases} \quad (38)$$

The level-dependent WDRC target gain is:

$$G_{target}(f, L_{in}) = \begin{cases} G_{CT}(f), & \text{if } L_{band}(f) \leq \text{CT}_{band}(f) \\ G_{CT}(f) - (L_{band}(f) - \text{CT}_{band}(f)) \left(1 - \frac{1}{\text{CR}_{loud}(f)}\right), & \text{if } L_{band}(f) > \text{CT}_{band}(f) \end{cases} \quad (39)$$

## 5. Demographic Adjustments:

$$G_{target}(f, L_{in}) \leftarrow G_{target}(f, L_{in}) + \Delta_{gender} + \Delta_{config} + \Delta_{exp} \quad (40)$$

where  $\Delta_{gender} = -1.5$  dB (female),  $\Delta_{config} = +3.0$  dB (unilateral), and  $\Delta_{exp} = -\min(6.0, 0.3 \cdot \max(0, \text{PTA}_{500,1k,2k} - 40))$  for new users.

**TABLE III. Effective Compression Ratios (Gain50 / Gain80) across A1-A7 Audiograms.**

| Profile | Formula | 250 Hz | 500 Hz | 1000 Hz | 2000 Hz | 4000 Hz | 8000 Hz |
|---------|---------|--------|--------|---------|---------|---------|---------|
| A1      | NAL-NL2 | -      | -      | 62.00   | 7.22    | 3.66    | 2.74    |
| A1      | Open-NL | -      | 1.00   | 2.02    | 2.84    | 2.60    | 8.45    |
| A2      | NAL-NL2 | -      | -      | 5.84    | 47.33   | -       | -       |
| A2      | Open-NL | -      | 5.59   | 2.71    | 6.41    | -       | -       |
| A3      | NAL-NL2 | 1.00   | -      | 8.37    | 3.75    | 3.03    | 2.42    |
| A3      | Open-NL | 1.00   | 1.00   | 2.31    | 2.94    | 2.94    | -       |
| A4      | NAL-NL2 | 1.00   | 1.00   | -       | 6.40    | 2.46    | 2.11    |
| A4      | Open-NL | 1.00   | 1.00   | 1.00    | -       | 2.96    | 1.64    |
| A5      | NAL-NL2 | 1.00   | 1.00   | 56.50   | 2.44    | 1.87    | 1.75    |
| A5      | Open-NL | 1.00   | 1.00   | 1.00    | -       | 2.17    | 2.10    |
| A6      | NAL-NL2 | 1.35   | 1.35   | 1.43    | 1.43    | 1.55    | 1.53    |
| A6      | Open-NL | 1.54   | 1.38   | 1.30    | 1.29    | 1.26    | 1.34    |
| A7      | NAL-NL2 | 1.00   | 1.00   | 1.00    | 1.00    | 1.00    | 1.00    |
| A7      | Open-NL | 1.00   | 1.00   | 1.00    | 1.00    | 1.00    | 1.00    |

## N. Stage 11: Acoustic Venting, Coupling Loss, and Receiver Saturation Limits

Real-Ear Aided Responses (REAR) are heavily influenced by acoustic coupling. Low-frequency leakage is modeled by log-frequency interpolation over anchor frequencies  $\mathbf{f}_{vent} = [250, 500, 1000, 2000, 4000, 8000]$  Hz:

$$V_{loss}(f) = \text{interp}_{\log_{10}}(f, \mathbf{f}_{vent}, \mathbf{v}_c) \quad (41)$$

where  $\mathbf{v}_c$  is the coupling-specific attenuation vector defined in Table S1.

**TABLE S1. Acoustic Coupling Real-Ear Insertion Loss Vectors ( $v_c$ , in dB).**

| Coupling<br>Config-<br>uration | 250 Hz | 500 Hz | 1000 Hz | 2000 Hz | 4000 Hz | 8000 Hz |
|--------------------------------|--------|--------|---------|---------|---------|---------|
| custom_occluded                | 0      | 0      | 0       | 0       | 0       | 0       |
| open_dome                      | -35    | -28    | -15     | -2      | 0       | 0       |
| tulip_dome                     | -25    | -18    | -5      | 0       | 0       | 0       |
| double_dome                    | -20    | -10    | 0       | 0       | 0       | 0       |
| vent_1mm_solld                 | -1     | 0      | 0       | 0       | 0       | 0       |
| vent_2mm_solld                 | -2     | 0      | 0       | 0       | 0       | 0       |
| vent_3mm_solld                 | -4     | 0      | 0       | 0       | 0       | 0       |
| vent_1mm_hollow                | -3     | 0      | 0       | 0       | 0       | 0       |
| vent_2mm_hollow                | -12    | -5     | -2      | 0       | 0       | 0       |
| vent_3mm_hollow                | -15    | -8     | -4      | 0       | 0       | 0       |

Conductive air-bone gaps are restored linearly with a 75% fraction:  $G_{cond}(f) = 0.75 \cdot \text{Loss}_{cond}(f)$  (Note: This 75% restoration fraction is a pragmatic engineering convention—adapted from clinical practice to prevent excessive output demands and MPO clipping; Johnson, 2013a—without direct empirical derivation from listener preference). To prevent active anti-phase cancellation demands and comb filtering, insertion gain is floored at  $V_{loss}(f) - 10$  dB:

$$G_{heuristic}(f, L_{in}) = \max(G_{target}(f, L_{in}) + 0.75 \cdot \text{Loss}_{cond}(f) + V_{loss}(f), V_{loss}(f) - 10) \quad (42)$$

Hardware receiver limits (MPO/SSPL90) are established to avoid severe saturation distortion:

$$\text{MPO}(f) = \min(120, 105 + 0.5 \cdot \max(0, \text{HTL}_{sn}(f) - 20) + \text{Loss}_{cond}(f)) \quad (43)$$

## O. Stage 12: Embedded Nelder-Mead Simplex Optimization & Physiological Loudness Ceilings

When `optimize = TRUE`, Open-NL adjusts the heuristic targets by minimizing an unconstrained multi-objective loss function via Nelder-Mead simplex search (`stats::optim`).

### Parameter Vector and Gain Formation

The optimization parameter vector is  $\delta = [\delta_{250}, \delta_{500}, \delta_{1000}, \delta_{2000}, \delta_{4000}, \delta_{8000}]^T \in \mathbb{R}^6$ . During evaluation, shifts are clamped:

$$\delta_{clamped,j} = \max(-60, \min(10, \delta_j)) \quad (44)$$

Candidate insertion gains  $\mathbf{G} \in [0, 80]$  dB are interpolated to calculation frequencies:

$$G(f) = \max(0, \min(80, G_{heuristic}(f) + \delta_{clamped}(f))) \quad (45)$$

### Loss Function Formulation

The Nelder-Mead solver minimizes:

$$\mathcal{L}(\boldsymbol{\delta}) = -100 \cdot \text{SII}_{desens}(\mathbf{G}) + P_{anchor} + P_{bounds} + P_{loud} + P_{spl} + P_{rough} + P_{order} + P_{cr} + P_{abg} \quad (46)$$

where  $\text{SII}_{desens}$  is the effective Speech Intelligibility Index calculated using the `johnson2011_smoothed` transfer function.

### Explicit Penalty Terms and Exact Weights ( $\lambda$ )

1. **Anchor Penalty** ( $\lambda_{anchor} = 0.1$ ):

$$P_{anchor} = 0.1 \sum_{j=1}^6 |\delta_j| \quad (47)$$

2. **Out-of-Bounds Penalty** ( $\lambda_{bounds} = 1000.0$ ):

$$P_{bounds} = 1000.0 \left[ \sum_{j=1}^6 \max(0, \delta_j - 30)^2 + \sum_{j=1}^6 \max(0, -\delta_j - 60)^2 \right] \quad (48)$$

3. **Physiological Loudness Ceiling Penalty** ( $\lambda_{loud} = 2000.0$ ):

$$P_{loud} = 2000.0 \cdot \max(0, \text{Sones}_{MG04}(\mathbf{G}) - \text{Cap}) \quad (49)$$

where  $\text{Sones}_{MG04}$  is the Moore & Glasberg (2004) specific-loudness integration computed via native C++. The dynamic U-shaped cap is interpolated across Pure Tone Average knots  $\mathbf{PTA}_{knots} = [10, 32.5, 52.5, 72.5, 90]$  dB HL from level-specific sone vectors:

- $L_{in} = 50$  dB SPL:  $\mathbf{K}_{50} = [1.5, 1.0, 0.8, 1.2, 1.2]$  sones
- $L_{in} = 65$  dB SPL:  $\mathbf{K}_{65} = [7.0, 4.5, 4.0, 6.5, 6.0]$  sones
- $L_{in} = 80$  dB SPL:  $\mathbf{K}_{80} = [20.0, 12.0, 10.0, 15.0, 14.0]$  sones

$$\text{Cap}_{base} = \text{interp}(\text{PTA}_{sn}, \mathbf{PTA}_{knots}, \mathbf{K}_{L_{in}}) \quad (50)$$

*Profile Adjustments:*

- Reverse-slope restriction ( $L_{in} \geq 75$  dB SPL and low-to-high threshold difference  $> 10$  dB):  $\text{Cap} = \text{Cap}_{base} - 0.10 \cdot (\overline{\text{HTL}}_{\leq 500} - \overline{\text{HTL}}_{\geq 4000})$ .
- Conductive air-bone gap adjustment:  $\text{Cap} \leftarrow \text{Cap} - 0.25 \cdot \text{PTA}_{ABG}$  (if  $L_{in} \geq 75$  dB SPL) or  $+0.10 \cdot \text{PTA}_{ABG}$  (if  $L_{in} < 75$  dB SPL).

4. **Broadband SPL Saturation Ceiling Penalty** ( $\lambda_{spl} = 2000.0$ ):

$$P_{spl} = 2000.0 \cdot \max(0, \text{SPL}_{aided} - 110.0) \quad (51)$$

5. **Spectral Roughness Penalty** ( $\lambda_{rough} = 0.5$ ):

$$P_{rough} = 0.5 \sum_{j=1}^5 (\delta_{j+1} - \delta_j)^2 \quad (52)$$

6. **Inter-Level Order Monotonicity Penalty** ( $\lambda_{order} = 2000.0$ ): Enforces  $G_{50} \geq G_{65}$  and  $G_{80} \leq G_{65}$ :

$$P_{order} = \begin{cases} 2000.0 \sum_{j=1}^6 \max(0, G_{65,j} - G_{50,j})^2, & \text{for } L_{in} = 50 \\ 2000.0 \sum_{j=1}^6 \max(0, G_{80,j} - G_{65,j})^2, & \text{for } L_{in} = 80 \end{cases} \quad (53)$$

7. **Compression Ratio Ceiling Penalty** ( $\lambda_{cr} = 200.0$ ):

$$P_{cr} = \begin{cases} 200.0 \sum_{j=1}^6 \max(0, (G_{50,j} - G_{65,j}) - \Delta_{max,j})^2, & \text{for } L_{in} = 50 \\ 200.0 \sum_{j=1}^6 \max(0, (G_{65,j} - G_{80,j}) - \Delta_{max,j})^2, & \text{for } L_{in} = 80 \end{cases} \quad (54)$$

where  $\Delta_{max,j} = 15.0 \cdot (1 - \text{ABG}_j / \max(0.001, \text{HTL}_j))$  for soft speech, bounding emergent compression ratios safely below 3.0:1 for sensorineural loss while enforcing linear amplification for conductive components. This 3.0:1 ceiling represents the best-supported parameter in the algorithm, firmly grounded in empirical psychoacoustic literature (Souza, 2002; Souza et al., 2006) demonstrating severe envelope flattening, loss of acoustic contrast, and speech-in-noise deficits for CRs  $> 3.0 : 1$ .

8. **Air-Bone Gap Excursion Penalty** ( $\lambda_{abg} = 1.0$ ):

$$P_{abg} = \begin{cases} 1.0 \sum_{j=1}^6 \max(0, \delta_j)^2, & \text{if } \exists j, \text{Loss}_j > 0 \\ 0, & \text{otherwise} \end{cases} \quad (55)$$

### Solver Controls and Convergence Tolerances

- **Algorithm:** Nelder-Mead Simplex via R's `stats::optim()`.
- **Iteration Ceiling:** `control = list(maxit = 800)`.
- **Convergence Tolerance:** Relative convergence tolerance `reltol = sqrt(.Machine$double.eps) \approx 1.49 \times 10^{-8}`.

- **Initial Simplex Seeding:** Seeding incorporates an audibility projection  $\delta_{start} = \min(20, \max(0, \text{target\_aided} - G_{heuristic}))$ , where  $\text{target\_aided} = \min(\text{UCL} - 5, \max(L_{in}, \text{HTL} + 10))$ . Soft inputs receive +3 dB shift; loud inputs receive  $\max(-10, \delta_{start} - 5)$  dB shift.
- **Multi-Start Strategy:** 3 restarts executed per input level with uniform random jitter  $\delta_{start} \leftarrow \delta_{start} + \mathcal{U}(-5, +5)$  to escape shallow local extrema and guarantee global convergence.

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